

IoT SMART REMINDER SYSTEM

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OVERVIEW

1. Project background and description

The Internet of Things Smart Reminder System is a personal assistant designed to enhance task management by using real-time environmental data such as geolocation to send time based triggers. The IoT Smart Reminder System uses multiple Raspberry Pis, a cloud database, and a mobile application to deliver timely reminders via audio and visual alerts to enhance time management by reminding the user of tasks they had previously mentioned, as well as suggesting the locations of objectives in proximity to the user's location.

2. Key Features

- Location Based Triggers: Track the users location via the mobile app to activate reminders based on geolocation.
- Environmental Monitoring: A second Raspberry Pi with Sense Hat to monitor temperature, humidity, and motion.
- Mobile Integration: Mobile app to allow users to set and receive reminders with real-time updates.
- Centralized Database: A cloud hosted database to store user preferences, reminders, and sensor data.

3. Specific exclusions from scope

Some components that we may need to purchase are but not limited to:

1. USB Microphone
2. Mini Speaker
3. LED Display

4. Timeline/Implementation Plan

1. **Week 4-5:**
 - Set up Raspberry Pi and Sense HAT.
 - Develop basic time-based reminders with LED outputs
 - Start building an iOS app using Swift.
2. **Week 5-6:**
 - Add environmental triggers using Sense HAT sensors.
 - Continue developing iOS app
 - Start implementing a cloud database
3. **Week 7-8:**
 - Implement mobile notifications via email/SMS.
 - Integrate a USB microphone for voice inputs.
4. **Week 9-10:**
 - Build and test
5. **Week 11-12:**
 - Finalize last components and get project ready for presentation